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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the derivative of the function.

$$y = 6xe^{-7x} - 3e^{x^8}$$

Use the Sum Rule to differentiate the expression. Evaluate each derivative.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(6xe^{-7x} - 3e^{x^8}) \\ &= \frac{d}{dx}(6xe^{-7x}) + \frac{d}{dx}(-3e^{x^8})\end{aligned}$$

Evaluate the derivative of the first term using the Product Rule. When finding the derivative of e^{-7x} , identify u and apply the rule $\frac{d}{dx}e^u = e^u \frac{du}{dx}$.

$$\frac{d}{dx}(6xe^{-7x}) = 6e^{-7x} - 42xe^{-7x}$$

Evaluate the derivative of the second term using the Chain Rule. If $y = f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

To represent the function $-3e^{x^8}$ in the form $y = f(u)$ and $u = g(x)$, let $y = -3e^u$ and let $u = x^8$.

Substitute $-3e^u$ for $f(u)$ and x^8 for $g(x)$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{d}{du}f(u) \cdot \frac{d}{dx}g(x) \\ &= \frac{d}{du}(-3e^u) \cdot \frac{d}{dx}(x^8)\end{aligned}$$

So, $y = -3e^u$ and $u = x^8$. First find $\frac{d}{du}(-3e^u)$.

$$\frac{d}{du}(-3e^u) = -3e^u$$

Now find $\frac{d}{dx}(x^8)$.

$$\frac{d}{dx}(x^8) = 8x^7$$

Now use the results to find the derivative of $-3e^{x^8}$. Evaluate $\frac{dy}{du}$ at $u = g(x)$. Recall that $\frac{dy}{du} = -3e^u$, $\frac{du}{dx} = 8x^7$, and $u = x^8$.

$$\frac{d}{dx}(-3e^{x^8}) = -24x^7 e^{x^8}$$

Now add the derivatives. Recall that $\frac{d}{dx}(6xe^{-7x}) = 6e^{-7x} - 42xe^{-7x}$ and $\frac{d}{dx}(-3e^{x^8}) = -24x^7 e^{x^8}$.

$$\begin{aligned}\frac{d}{dx}(6xe^{-7x} - 3e^{x^8}) &= \frac{d}{dx}(6xe^{-7x}) + \frac{d}{dx}(-3e^{x^8}) \\ &= 6e^{-7x} - 42xe^{-7x} - 24x^7 e^{x^8}\end{aligned}$$

So, the derivative of $6xe^{-7x} - 3e^{x^8}$ is $6e^{-7x} - 42xe^{-7x} - 24x^7 e^{x^8}$.